

Mouse dLGN Receives Functional Input from a Diverse Population of Retinal Ganglion Cells with Limited Convergence

Highlights

- The majority of functionally identified RGCs types project to dLGN
- Factorization of the dLGN population response reveals diverse components
- Responses of dLGN cells can be modeled as combination of, on average, 5 RGC types
- Among those, 2 RGC types have the most dominant functional impact

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In Brief

Román Rosón, Bauer, et al. show that most functionally identified retinal ganglion cell (RGC) types project to the dLGN of thalamus. A linear feedforward model accounts for dLGN responses using input from ~5 RGC types, of which 2 are dominant.



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SUMMARY

Mouse vision is based on the parallel output of more than 30 functional types of retinal ganglion cells (RGCs). Little is known about how representations of visual information change between retina and dorsolateral geniculate nucleus (dLGN) of the thalamus, the main relay between retina and cortex. Here, we functionally characterized responses of retrogradely labeled dLGN-projecting RGCs and dLGN neurons to the same set of visual stimuli. We found that many of the previously identified functional RGC types innervate dLGN, which maintained a high degree of functional diversity. Using a linear model to assess functional connectivity between RGC types and dLGN neurons, we found that responses of dLGN neurons could be predicted as linear combination of inputs from on average five RGC types, but only two of those had the strongest functional impact. Thus, mouse dLGN receives functional input from a diverse population of RGC types with limited convergence.

INTRODUCTION

In the mammalian retina, the photoreceptor signal is decomposed into multiple parallel channels (Euler et al., 2014; Masland, 2012), carried to the brain by more than 30 types of retinal ganglion cells (RGCs) (Baden et al., 2016; Sanes and Masland, 2015). Each type of RGC extracts specific features of the visual environment, which are projected via the optic nerve to more

than 50 retino-recipient areas in the brain (Martersteck et al., 2017; Morin and Studholme, 2014). Among those, a key center transmitting information to the primary visual cortex (V1) is the dorsolateral geniculate nucleus (dLGN) of the thalamus.

Retino-geniculate information transmission has been studied extensively in cats and monkeys, where the majority of dLGN neurons seems to be driven by only few (1–3) dominant RGCs (Cleland et al., 1971; Usrey et al., 1999). This dominant input can evoke such strong excitatory postsynaptic potentials (EPSPs)—so-called “S-potentials”—that they can be picked up by extracellular recordings. Consistent with a low degree of convergence, the S-potentials of a dLGN neuron and its spiking output have closely matching receptive fields (RFs) in terms of location, center-surround organization, and size (Hubel and Wiesel, 1961; Kaplan et al., 1987; Sincich et al., 2007). Besides these dominant inputs, individual cat dLGN cells receive additional weaker inputs, as shown by *in vivo* electrophysiology (Cleland et al., 1971; Mastronarde, 1987, 1992; Usrey et al., 1999) and inferred from ultrastructural assessment of synaptic bouton placement and number (Hamos et al., 1987). Given the strict spatial layering of the dLGN, in particular in primates, it has generally been assumed that, in these species, the inputs into individual dLGN neurons arise from RGCs of similar functional type, although at least 13 anatomical types of dLGN-projecting RGCs have been identified in monkeys (Dacey et al., 2003).

In the mouse, recent anatomical studies have started to explore retino-geniculate connectivity and revealed a more complex picture. Mono-transsynaptic rabies virus tracing of inputs to individual dLGN neurons demonstrated that mouse dLGN neurons can be divided into two groups based on the pattern of their retinal inputs (Rompani et al., 2017): although some dLGN neurons received inputs from mostly a single RGC type (“relay mode”), others showed a high degree of convergence, with inputs being composed of up to several dozen RGCs of different

types (“combination mode”). A high degree of retino-geniculate convergence is further supported by recent ultrastructural studies of retinal afferents and their thalamic relay cell targets (Hammer et al., 2015; Morgan et al., 2016).

Given the functional diversity of mouse RGCs (Baden et al., 2016; Farrow and Masland, 2011; Sanes and Masland, 2015), whose properties might vary with retinal region (Bleckert et al., 2014; Joesch and Meister, 2016; Nath and Schwartz, 2016; Warwick et al., 2018), and complex input of RGCs to dLGN neurons (Ellis et al., 2016; Hammer et al., 2015; Liang et al., 2018; Morgan et al., 2016; Rompani et al., 2017), one would expect an at least similarly rich functional representation in mouse dLGN. In contrast, the majority of mouse dLGN neurons has been reported to have circularly symmetric RFs (Piscopo et al., 2013) and is believed to perform linear spatial summation (Denman and Contreras, 2016; Grubb and Thompson, 2003), similar to primates (Grubb and Thompson, 2003). However, mouse dLGN also contains neurons with more complex and diverse response properties: orientation-selective (OS) and direction-selective (DS) cells (Cruz-Martín et al., 2014; Marshel et al., 2012; Piscopo et al., 2013; Zhao et al., 2013), “suppressed-by-contrast” neurons, potentially signaling uniformity of the visual field (Piscopo et al., 2013), as well as a heterogeneous population of cells with long latencies and responses to both light on- and offset (Piscopo et al., 2013). It is currently unknown to which degree these response properties are inherited from the innervating retinal afferents (Cruz-Martín et al., 2014; Liang et al., 2018) or emerge *de novo* in dLGN by a combination of converging retinal inputs and dLGN-intrinsic computations (Liang et al., 2018; Marshel et al., 2012).

Here, we sought to determine how the visual representation in mouse dLGN arises from retinal inputs. We show that many previously identified functional RGC types (Baden et al., 2016) innervate the dLGN, where, in accordance with this richness of input, the population response consists of diverse fundamental components. The dLGN responses in response to our simple, full-field stimuli, and at the temporal resolution of calcium signals, can be modeled by a linear combination of on average five RGC types, among which two have the strongest functional impact. We conclude that mouse dLGN neurons receive functional input from multiple RGC types and relay these diverse retinal representations to the cortex with limited convergence.

RESULTS

Retrograde Viral Tracing to Functionally Characterize dLGN-Projecting RGCs

To identify dLGN-projecting (dLGN-p) RGCs, we injected a Cre-encoding retrograde herpes simplex virus 1 (LT HSV-hEF1 α -cre; Neve, 2012) into the dLGN of a transgenic reporter mouse line with a floxed genetically encoded Ca²⁺ indicator (GCaMP6f; Chen et al., 2013; Madisen et al., 2015). After transducing the axon terminals (Antinone and Smith, 2010; McGavern and Kang, 2011) in the dLGN, the virus was retrogradely transported to the soma, where it triggered the expression of Cre-recombinase and, subsequently, the Cre-dependent expression of GCaMP6f. Because the virus does not spread *trans-synapti-*

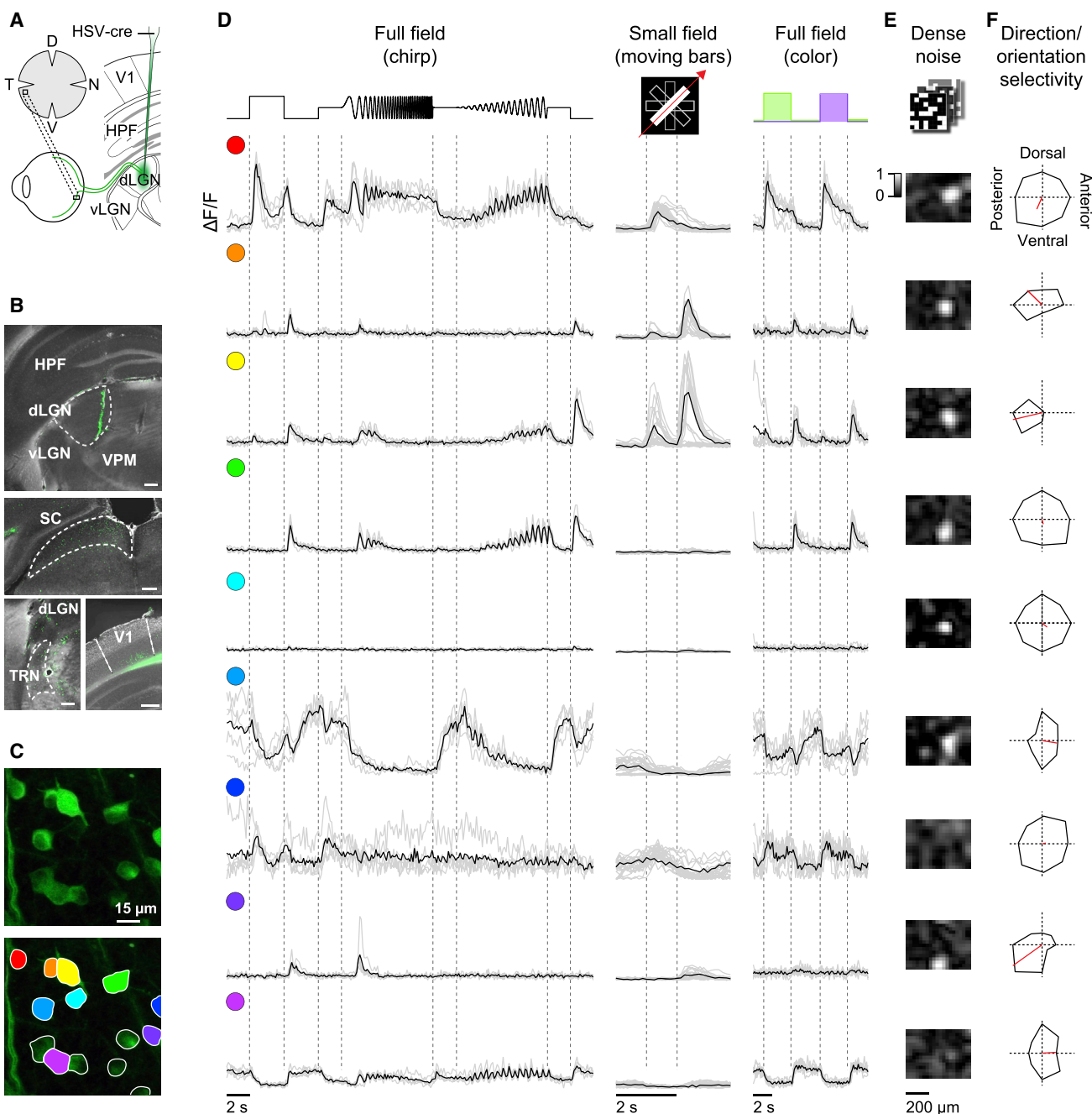
cally, it only labeled cells with afferents in the dLGN. This enabled us to identify only the subset of dLGN-p RGCs in the retina (Figure 1A).

We histologically confirmed the target location of the virus injection, as well as its limited diffusion (Figures 1B, top, and S1). Given our dorsal approach, the HSV virus was injected into both shell and core regions of dLGN, but due to the sheer volume difference between these regions, it is likely that more terminals were transduced in the dLGN core. In line with earlier studies, we found retrogradely labeled neurons in additional dLGN-projecting structures, including the superior colliculus (Figure 1B, center), the thalamic reticular nucleus, and the deep layers of primary visual cortex (Figure 1B, bottom; Guillery and Sherman, 2002; Harting et al., 1991).

We then used two-photon Ca²⁺ imaging to measure the light-evoked responses of the dLGN-p RGCs (Figures 1C–1F). We probed their response properties across the whole retina with a stimulus set used in a previous RGC classification study (Baden et al., 2016). As the labeled RGCs were sparse (Figure 1C, top), we analyzed their Ca²⁺ responses using manually drawn regions of interest (ROIs) (Figure 1C, bottom). In total, we identified 581 virus-labeled RGCs from 4 mice (contralateral eye only) as ROIs, with a range of 1–25 ROIs per field (median = 6). As expected from earlier work, we found ON- and OFF-RGCs (without any direction or orientation preference; Huberman et al., 2008) as well as direction-selective RGCs (Cruz-Martín et al., 2014; Ellis et al., 2016; Rivlin-Etzion et al., 2011). In addition, we found dLGN-p RGCs that, for example, responded differently to local and full-field stimuli, showed preference to higher or lower frequency stimulation, or were suppressed by frequency and contrast (Figures 1D–1F).

The Majority of Functional RGC Types Project to dLGN

We next asked which of the previously characterized functional mouse RGC types (Baden et al., 2016) project to the dLGN. We used the functional RGC clusters obtained previously (henceforth “RGC-all clusters”) and identified, for each retrogradely labeled dLGN-p RGC, the RGC-all cluster with the best matching response properties (Figure S2). To account for the differences in Ca²⁺ indicators between the two studies, we first de-convolved the Ca²⁺ signals of the GCaMP6f dataset and the published OGB-1 dataset (Baden et al., 2016), using each indicator’s Ca²⁺ kernels, calculated as average thresholded Ca²⁺ peak events to the white noise stimulus of multiple ROIs (Figure S3; see STAR Methods). For both the “chirp” stimulus and the moving bar stimulus, we correlated the trial-averaged responses of each dLGN-p RGC to the mean response of each RGC-all cluster. The correlation coefficients were then weighted by a quality index (*Q*), which is a measure of signal-to-noise ratio and reproducibility of the responses, and averaged. This weighted mean correlation (“match index” [*Mi*]) was then used to assign each dLGN-p RGC to the best-matching RGC-all cluster, i.e., the cluster with the highest *Mi* (for statistics of *Mi*, see Figure S4). For both individual example cells (Figure 2A), and across the population of dLGN-p RGCs, the assignment worked well (median *Mi* = 0.62). Accordingly, population mean responses of dLGN-p RGCs assigned to the same



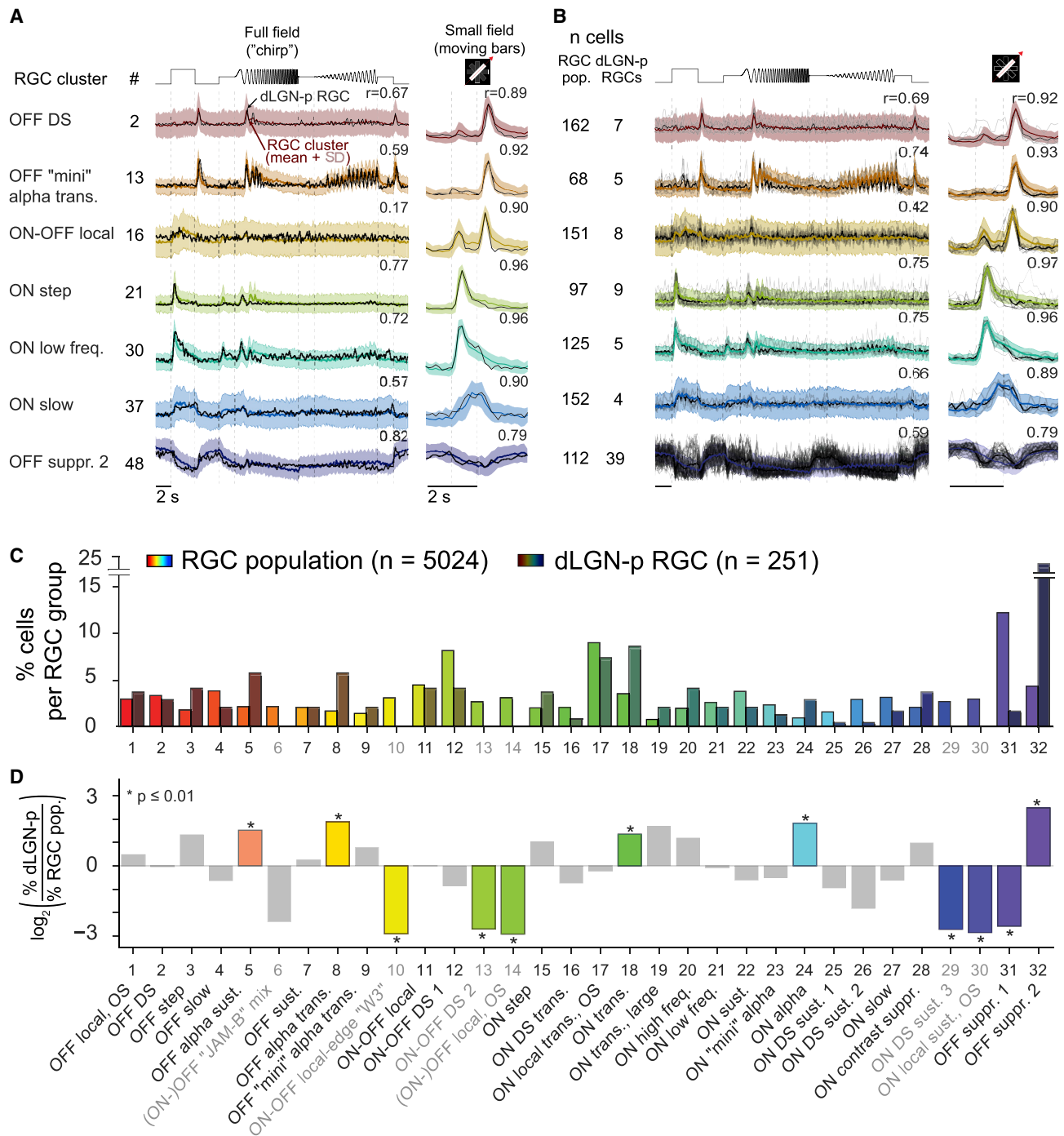


Figure 2. Cluster Assignments

(A) Responses of selected RGC-all clusters (colored line, mean; shaded area, ± 1 SD) with mean responses of assigned example dLGN-p RGCs (black) to chirp (left) and bar stimulus (right). Numbers indicate correlation coefficients.

(B) Same, for all assigned dLGN-p RGCs in the corresponding clusters (gray, responses of single dLGN-p RGCs; black, mean).

(C) Distribution of cells per RGC-all group from [Baden et al. \(2016\)](#) versus cells per dLGN-p group.

(D) Comparison of cell-per-group percentages as $\log_2$ ratio (%dLGN-p RGCs/%RGC-all). Significant differences in cell proportions ($p < 0.01$; binomial test) are marked as colored bars and with asterisks.

See also [Figures S3](#), [S4](#), and [S5](#).

cluster agreed well with the population mean responses of their RGC-all cluster (Figure 2B).

We then determined which RGC types were over- or underrepresented in the dLGN-p RGC population compared to the complete RGC population. Following Baden et al. (2016), we grouped the 49 RGC-all clusters into 32 groups, where each group represents an RGC type based on domain knowledge not only about functional features but also about soma size, RF size, and direction and orientation selectivity (Figure 2C; see STAR Methods; see Figure S5 for same analysis without grouping). This analysis revealed that 75% of RGC-all groups were assigned dLGN-p RGCs (24/32 groups with at least two cells), suggesting that the majority of functional RGC types projects to the dLGN.

Several RGC types were systematically over- or underrepresented in the dLGN-p population (Figures 2C and 2D; binomial test, $p < 0.01$): overrepresented were “OFF-suppressed 2” cells (Figures 2C and 2D: group [G] 32) and all classical alpha cells (Figures 2C and 2D), including OFF alpha sustained (G_5), OFF alpha transient (G_8), and ON alpha (G_{24}), as well as an ON transient RGC (G_{18}). Underrepresented, in turn, were the ON-OFF local-edge “W3” (G_{10}) and several orientation- and direction-selective RGCs (i.e., ON-OFF DS 2 [G_{13}], [ON-]OFF local OS [G_{14}], ON DS sust. 3 [G_{20}], and ON local sustained OS [G_{30}]). Other groups contributed roughly the same percentage to the dLGN-p cells as to the total RGC population (see Discussion). These results indicate that dLGN receives diverse, parallel input from many functional RGC groups, with a striking overrepresentation of alpha and of one group of suppressed-by-contrast (SbC) RGCs.

The dLGN Population Response Can Be Factorized into Many Components

Having established that many functional RGC types provide input to mouse dLGN, we next asked how this diversity is reflected in the dLGN population. We performed extracellular single-unit recordings with linear silicon probes of geniculate neurons in head-fixed mice, using the same dorsal approach as for the virus injections (Figures 3A and 3B). We verified the recording sites by histological reconstruction of the electrode tract (Figure 3B) and the characteristic progression of retinotopy along the electrode channels (Figures 3C and S6; see also Piscopo et al., 2013). Given our dorsal approach to dLGN, the recorded neurons likely come from both the core and shell region of dLGN, although volume differences likely bias our results toward properties of the dLGN core. RF centers of the recorded neurons were approximately between -20° and $+50^\circ$ elevation and -10° and $+100^\circ$ azimuth (Figure S6). We presented the same full-field chirp stimulus as in the retinal experiments. To assess the stability of the dLGN recordings and the consistency of our spike sorting, we flanked the chirp by presentations of drifting gratings with varying orientation and temporal and spatial frequency (Figure S7).

Chirp responses of dLGN neurons were surprisingly diverse: the cells not only displayed the “standard” transient and sustained ON-OFF responses or were suppressed by contrast but also differed in their temporal frequency or contrast preferences as well as their response kinetics (Figure 3D). Some of the cells even displayed slow ramping responses (Figures 3D4 and 3D5). To ensure that this response diversity did not result from poor unit isolation during spike sorting, we considered only units

with firing rates >1 spike/s, which were stable across time, had a clean refractory period, and a distinct extracellular spike waveform (Figures 3E and 3F). This diversity is likely not related to influences of locomotion (Figure S8).

To quantitatively assess the degree of diversity present in the dLGN neuron population, we decomposed the single-unit responses to the chirp stimulus into “response components” using non-negative matrix factorization (NNMF) (Lee and Seung, 1999). Individual neuron responses can then be reconstructed as a weighted sum of these elementary response components (Figures 4A and S2). Interestingly, these components can be well correlated to the responses of our dLGN-p RGCs, supporting the NNMF approach for analyzing the diversity of geniculate representations. For example, decomposing our dLGN population response into only two fundamental components (ranks) revealed components resembling responses with ON and OFF features; increasing the rank number to four added temporal diversity with transient and sustained features, and increasing rank number further led to additional contrast-suppressed features (Figure 4B). To determine the mathematically optimal number of components, we used cross-validation (Williams et al., 2018; Figure 4C), where we repeatedly performed the factorization on different training sets for an increasing number of ranks (Figure 4C) and evaluated the mean squared errors (MSEs) of the NNMF model on validation sets not used for factorization (STAR Methods). For a conservative estimate, we chose the model with 5% more error than the minimal error, resulting in 29 components (Figures 4D and 4E). This is indicative of a large diversity in the dLGN population response.

Besides this rich representation of luminance steps, temporal frequencies, and changes in contrast in the overall dLGN population, 82/443 (18.5%) dLGN neurons only displayed weakly modulated responses to the chirp stimulus, despite having robust and consistent responses to the drifting gratings presented before and afterward (Figure S7). This response behavior is consistent with these dLGN neurons preferring local variations in luminance instead of uniform full-field stimuli, as is the case for $\sim 30\%$ of RGCs (Baden et al., 2016; their Figure 2b and extended data Figure 10), suggesting that the overall diversity of dLGN responses may be larger than reported here. Together, we conclude that the response diversity observed at the level of dLGN-p RGCs is also present in dLGN.

Modeling dLGN Responses as Linear Combinations of RGC Inputs

Next, we combined the dLGN-p RGC dataset (Figure 2) and the dLGN dataset (Figure 3) to study how the dLGN responses are computed from the retinal output. We first accounted for the differences in recording methods by convolving the dLGN spiking responses with the OGB-1 Ca^{2+} indicator kernel (Figures S3 and S9). We then used a parsimonious, linear model, constrained to have non-negative weights, to predict dLGN responses—on the temporal scale of Ca^{2+} signals—as a sum of weighted RGC inputs (Figures 5A and S2). For prediction, we used the RGC-all cluster means (Baden et al., 2016) that were assigned at least two dLGN-p cells. Each dLGN recording consisted of multiple trials (stimulus repetitions; $n = 10\text{--}30$), which were divided into training and test sets (50%/50%). The model was evaluated

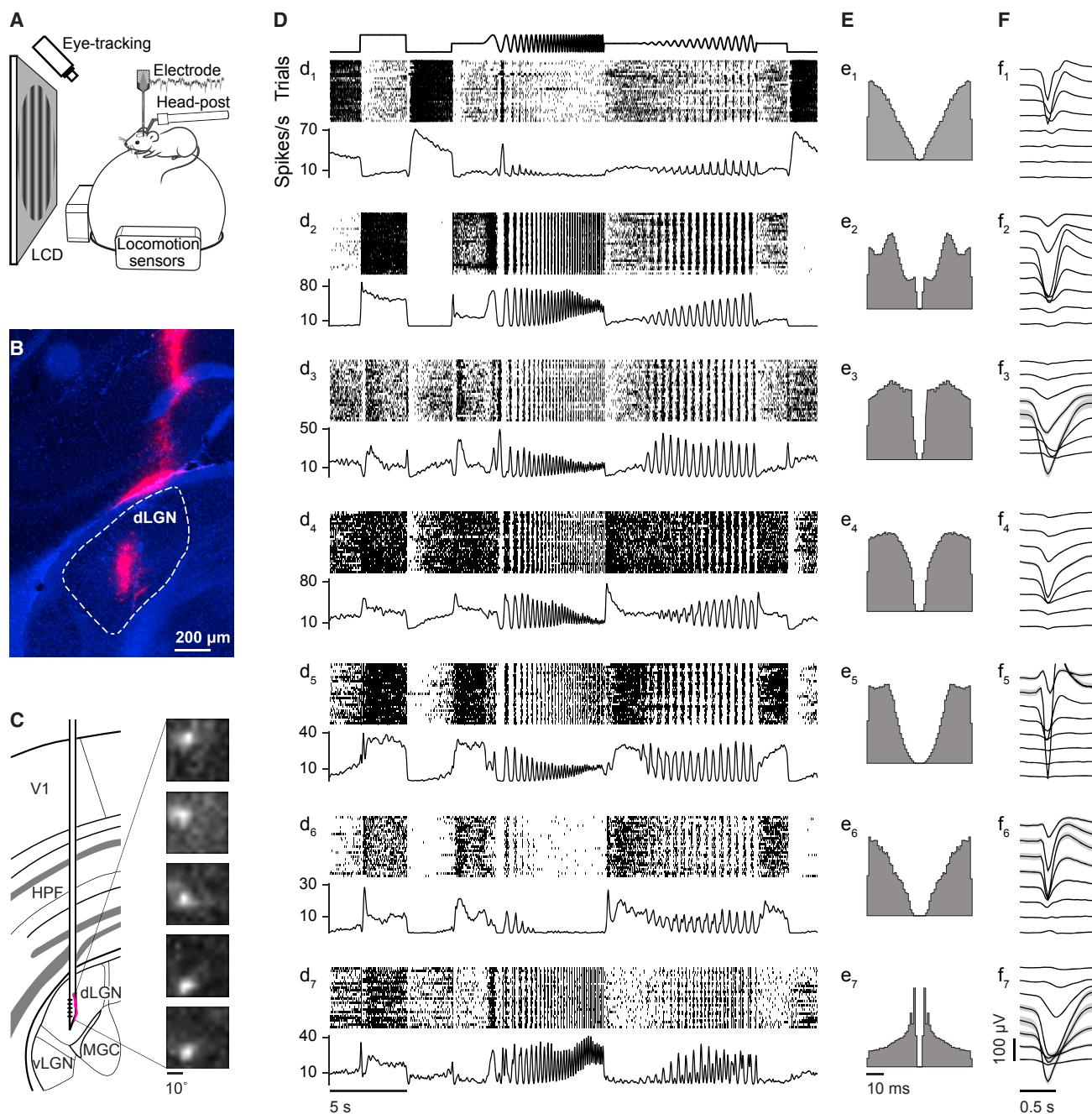


Figure 3. Functional Characterization of dLGN Neurons

(A) Schematic of experimental setup for extracellular dLGN recordings.

(B) Reconstruction of the electrode track in dLGN (coronal section; dLGN outlined in white; blue: DAPI; red: DiD coating the electrode).

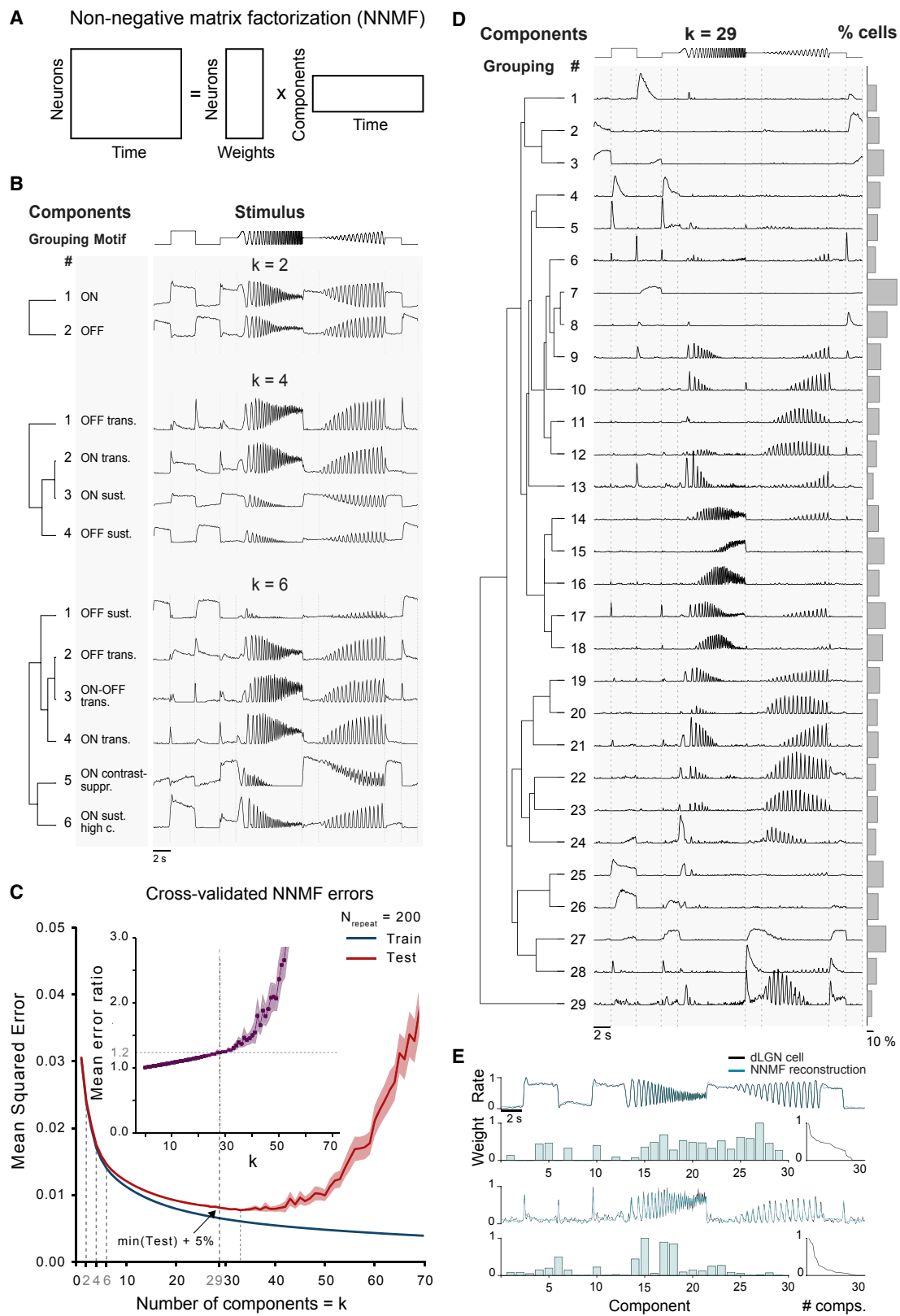
(C) Schematic of recording site with RFs mapped for several electrode channels (dorsal to ventral), showing the retinotopic progression in elevation typical of mouse dLGN.

(D–F) Spike raster plots (top) and spike density function (bottom) of 7 exemplary dLGN neurons in response to the chirp stimulus (D), their autocorrelograms (E), and spike waveforms in 5 selected channels of the 32-channel probe (F).

MGC, medial geniculate complex. See also [Figures S6, S7, and S8](#).

using repeated random sub-sampling cross-validation with 1,000 repetitions ([STAR Methods](#)). The reported weights represent mean values across the repeats.

With notable exception of the transient response changes ([Figure S9](#)), this simple excitation-only linear feedforward model successfully reconstructed responses of most dLGN neurons,



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including transient ON, sustained ON and OFF, and SbC cells (Figure 5B). Typically, the responses of dLGN neurons could be predicted using only a few non-zero weights (weight > 0.001; Figure 5B1–3); some cells were predicted by a combination of one dominant and few weaker RGC inputs (Figure 5B2), and even others were best modeled with responses from many RGC clusters (Figure 5B4). Across dLGN neurons, prediction quality was high (average correlation: 0.64; average RMSE: 0.17; Figure 5C).

Using this model, we next asked how many RGC types provided input to mouse dLGN neurons and found that the answer directly depends on the threshold used for considering inputs as functionally relevant (Figure 5D). Choosing a low weight threshold, the responses of dLGN neurons ($n = 814$) could be predicted by the combined input of on average 5 RGC groups (Figure 5D, left inset; mean = 5.33; median = 5.19; range = 2–11; middle 90% range = 3–8). Choosing a more conservative threshold for functionally relevant inputs, in turn, identified on average 2 most dominant RGC groups (Figure 5D, right inset; mean = 2.01). Together, these results indicate that the responses of many dLGN neurons, at least on the temporal scale of Ca^{2+} transients and for the presented set of visual stimuli, can be explained to a large extent by excitatory feedforward processing with limited convergence.

We also tested other, more flexible models, allowing also negative weights and an exponential non-linearity. Although these yielded small improvements in terms of prediction quality (~15%), they predicted convergence of dozens of RGC types, all with very low relative weights. Equalized input from many types without a dominant one is incompatible with results from *in vivo* recordings correlating spiking activity of RGC-dLGN neuron pairs (Carandini et al., 2007; Cleland et al., 1971; Hubel and Wiesel, 1961; Mastronarde, 1992; Rathbun et al., 2016; Usrey et al., 1999) and *in vitro* slice recordings (Chen and Regehr, 2000; Litvina and Chen, 2017), according to which some RGC inputs are clearly more dominant than others (reviewed in Alonso et al., 2006).

Finally, we explored which of the previously identified dLGN-p RGC groups were used for prediction in the model (Figure 5E). For the liberal weight threshold of 0.001, we found that the majority (24/32) of these groups were used by the model (Figure 5E, top), with some of them contributing more frequently than others. Interestingly, some of the groups contributing most strongly belonged to those RGC groups that were significantly overrepresented among the dLGN-p RGCs (Figure S10), although informa-

tion about the relative frequency of the projections was not used for inference. As expected, imposing a more conservative weight threshold resulted in an even more pronounced pattern of often-used RGC groups (Figure 5E, bottom), although their relative weights tended to equalize. Together, our modeling approach suggests that these RGC types have a significant functional role in the processing of visual information along the retinogeniculate pathway.

DISCUSSION

We functionally characterized the population of dLGN-projecting RGCs as well as dLGN neurons and provide a quantitative account of the functional connectivity between RGC types and dLGN neurons, for simple, full-field stimuli, and at the temporal resolution of Ca^{2+} signals. We present three main findings. First, combining a retrograde viral transduction approach and two-photon Ca^{2+} imaging of RGC responses, we show that the majority of previously functionally identified RGC types project to dLGN. Second, decomposing dLGN responses into their elementary response components revealed a rich diversity of geniculate visual representations, similar to that of RGCs. Third, we demonstrate that the responses of individual dLGN cells can be modeled as a linear combination of responses of on average 5 RGC types, among them 2 with the most dominant functional impact.

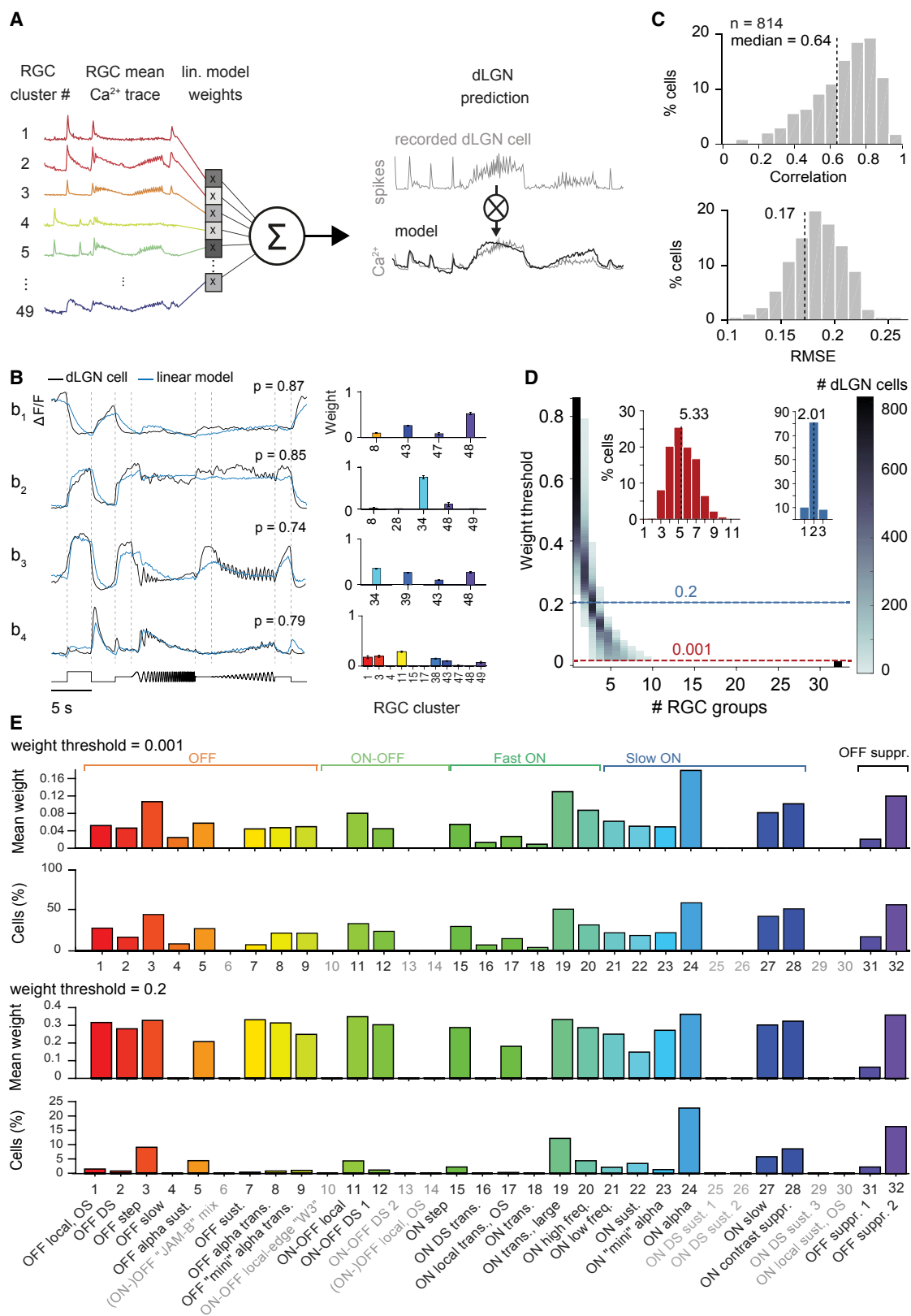
Retrograde Labeling of dLGN-Projecting RGCs

Retrograde viral tracing with HSV-cre combined with two-photon Ca^{2+} imaging of RGCs proved to be a suitable tool to determine functional properties of the dLGN-p RGCs. HSV is known to have a strictly synaptic uptake mechanism (Antinone and Smith, 2010; McGavern and Kang, 2011), which prevents infection of axons passing nearby and not synapsing within the dLGN. This is an issue with commonly used retrograde tracers, such as DiO/Dil, horseradish peroxidase, fluorophore-conjugated latex microspheres, or cholera toxin (Ellis et al., 2016). In addition, our observation that GCaMP6f expression in the retina was limited to spatially confined regions is consistent with the interpretation that HSV-infected neurons were restricted to the retinotopically corresponding region in dLGN instead of labeling passing RGC axons, which would likely have resulted in a larger spread of RGC locations.

To date, there are no data available supporting or arguing against an RGC type bias in HSV uptake. We think, however,

Figure 4. Decomposition of the dLGN Population Response

(A) Schematic of the non-negative matrix factorization (NNMF), used to extract a set of time-varying response components, from which neuronal responses can be reconstructed as a weighted sum.
 (B) Example NNMF models with $k = 2, 4$, and 6 components. Top: a model with two components clearly factorizes the data into an ON and OFF component. Middle: allowing four components introduces ON-transient and OFF-transient features in addition to sustained ON and OFF features. Bottom: with six components, components reflect ON-OFF transient as well as contrast suppressed features.
 (C) Mean squared errors (MSEs) as a function of model components k both for training (blue) and test (red) sets in NNMF model cross-validation ($n_{\text{repeats}} = 200$; lines, mean; shaded areas, CI_{95}). NNMF cross-validation was used to determine a conservative estimate of the optimal number of components as the location of the minimum mean $\text{MSE}_{\text{test}} + 5\%$ ($k = 29$). Inset: MSE ratio ($\text{MSE}_{\text{test}}/\text{MSE}_{\text{train}}$) per rank is shown (line: mean; shaded area: CI_{95}).
 (D) dLGN response components as computed by NNMF (middle), organized according to a hierarchical cluster tree with an optimized leaf order (left), and percentage of neurons with weight of respective component > 0 (right).
 (E) Top: examples of dLGN cell responses (black) and their NNMF reconstruction (blue). Bottom: weights used for reconstruction for each component (left), sorted in descending order (right).



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that viral tropism is likely not a major confound. First, we could retrogradely transfect the majority of functional RGC types, which is inconsistent with strong viral tropism. Second, even if there were biases, the amplification step of utilizing Cre to unlock transgene expression likely reduces potential differences, because only few viral particles retrogradely transported to the RGC soma suffice for robust cre-mediated GCaMP6f expression. Third, where direct comparisons with anatomical or genetic cell types are possible (see below and Table S1), our retrograde tracing experiments provide congruent results. Fourth, our results are overall in agreement with a previous retrograde labeling study (Ellis et al., 2016) using G-deleted rabies virus, indicating that the particular retrograde virus employed in our study does not grossly influence results about over- and underrepresentation of certain RGC types.

Functional Classification of dLGN-Projecting RGCs

The functional classification of dLGN-projecting RGCs, which was based on the functional classification obtained for the entire RGC population (Baden et al., 2016), revealed that around 75% of all functional RGC types innervate the dLGN. This high number of types is in line with a previous retrograde labeling study, which reported that, of the 88% of all RGCs projecting to the superior colliculus (SC), ~80% also send an axon collateral to dLGN (Ellis et al., 2016). Moreover, ON alpha RGCs and OFF transient alpha RGCs (Krieger et al., 2017; van Wyk et al., 2009) could be found more frequently among the dLGN-p cells than in the general RGC population. In agreement with this, ON-sustained alpha RGCs were shown to target the core region of the dLGN (Brown et al., 2010; Ecker et al., 2010; Hattar et al., 2006) and also OFF transient alpha RGCs project to dLGN (for a comparison to the literature, see Table S1). SbC RGCs were also overrepresented in our dLGN-p dataset. They typically exhibit a high baseline firing rate that is suppressed by light stimulation (Jacoby and Schwartz, 2018)—a type of response that has been recorded all along the mouse retino-geniculo-cortical pathway (Niell and Stryker, 2010; Piscopo et al., 2013; Tien et al., 2015), suggesting a dedicated early visual pathway signaling uniformity of the visual field (Levick, 1967; Masland and Martin, 2007; Sivyer et al., 2010) and/or controlling contrast gain (Troy et al., 1989).

Certain RGC types, like W3 RGCs or sustained ON DS RGCs, were underrepresented among dLGN-p cells, in agreement with the literature (Dhande et al., 2013; Kim et al., 2010; Yonehara et al., 2008, 2009; Zhang et al., 2012). In particular, the absence of dLGN-p cells among the W3 RGCs resonates well with find-

ings by Ellis et al. (2016) that ON RGCs with transient responses to small-diameter stimuli were more frequent among SC- than dLGN-projecting RGCs. In two cases, our data seem at odds with the previous literature: none of the dLGN-p RGCs in our study were classified as JAM-B cells (Kim et al., 2008), which are known to project to dLGN. This is likely because JAM-B cells barely respond to our stimuli set, and is consistent with Baden et al. (2016), where some RGC types, including JAM-Bs, did neither respond strongly to full-field chirps nor moving bars, and therefore, our clustering method had assigned these to a “mixed” RGC group. Also, OFF-sustained alpha RGCs were overrepresented among dLGN-p cells, although RGCs marked in the transgenic W7-line thought to correspond to OFF-sustained alpha RGCs have been shown not to project to dLGN (Kim et al., 2010). Interestingly, there was a cluster in the OFF-sustained group that was not assigned any dLGN-p cells, possibly indicating that this subgroup corresponds to W7A. Alternatively, W7A RGCs are a different RGC type—in this case, according to our data, all four alpha RGC types (ON-sustained, OFF-sustained, ON-transient, OFF-transient; Krieger et al., 2017) would project to dLGN, splitting the visual signal on the way to dLGN into four channels arranged symmetrically with respect to polarity and kinetics (Krieger et al., 2017; Pang et al., 2003; van Wyk et al., 2009).

Components of the dLGN Population Response

Consistent with our result of most functional RGC types projecting to dLGN and previous studies demonstrating rich retinal input to dLGN (Ellis et al., 2016; Liang et al., 2018), we found that the dLGN population responses can be factored into ~29 components. The specific number certainly depends on exactly which method is used for determining the optimal rank of the NMF. We tested several of these methods (Akaike information criterion [AIC], Bayesian information criterion [BIC], randomization, and cross-validation) and found none of them to suggest numbers as low as the classical notion of 3 parallel pathways (reviewed in Kerschensteiner and Guido, 2017). Instead, the high number of NMF components is—at least in mice—consistent with a larger diversity of dLGN inputs than commonly appreciated. These diverse inputs, in turn, will likely mediate or contribute to diverse visual features being represented in dLGN. This interpretation is supported by recent studies reporting “non-classical” responses in rodent dLGN (Cruz-Martín et al., 2014; Howarth et al., 2014; Marshel et al., 2012; Piscopo et al., 2013; Scholl et al., 2013; Zhao et al., 2013), rabbit dLGN

Figure 5. Functional Modeling of Retinogeniculate Convergence

- (A) Illustration of the linear model, predicting responses of a dLGN neuron, convolved with a Ca^{2+} indicator kernel, as a linear combination of weighted dLGN-p RGC cluster inputs.
- (B) Example responses of dLGN neurons (black) and predictions by the linear model (blue), along with the relative weights > 0.001 of RGC clusters used for modeling of dLGN responses.
- (C) Prediction quality. Top: correlation between actual response and predicted response for all dLGN neurons is shown ($n = 814$). Bottom: distribution of root mean squared errors is shown.
- (D) Distribution of the number of RGC groups used for reconstruction of dLGN neuron responses, shown as a function of weight threshold to consider when an input is functionally relevant. (Insets) Histograms of the percentage of dLGN cells as a function of number of RGC groups used for modeling are shown, considering only weights >0.001 (left) or >0.2 (right) as functionally relevant.
- (E) Distribution of RGC groups from Baden et al. (2016) used for reconstructing the population of dLGN cell responses, shown as mean weight and percentage of dLGN cells (top: weight threshold = 0.001; bottom: weight threshold = 0.2). RGC types not projecting to dLGN are greyed out.
- See also Figures S9 and S10.

(Hei et al., 2014), and the koniocellular layers of primate dLGN (Cheong et al., 2013; White et al., 2001; Zeater et al., 2015), such as direction and orientation selectivity and binocularity.

Recombination of RGC Channels in dLGN

In previous work, the question of retinogeniculate connectivity has been addressed from two different angles. Most studies focused on the absolute number and strength of RGC inputs to single dLGN neurons, whereas fewer studies also assessed convergence of retinogeniculate inputs in terms of RGC types.

Estimates of the absolute number of individual RGCs providing input to a single dLGN relay cell differ widely between studies (1 to ~90), with the discrepancy likely arising from differences in species, age, and method. Both *in vitro* and *in vivo* physiology studies concluded that dLGN neurons typically receive input from 1–3 dominant RGCs, with the possibility of additional, weaker inputs in cats and rodents (Chen and Regehr, 2000; Cleland et al., 1971; Jaubert-Miazza et al., 2005; Litvina and Chen, 2017; Mastronarde, 1992; Reid and Usrey, 2004; Usrey et al., 1999; Weyand, 2016; Ziburkus and Guido, 2006). For example, in parasagittal dLGN slices of adult mice, Litvina and Chen (2017) reported three dominantly driving inputs, with an average total number of 10 RGC inputs per dLGN relay cell. Recent anatomical studies using electron microscopy (EM) reconstructions (Hammer et al., 2015; Morgan et al., 2016) and mono-transsynaptic rabies virus tracing (Rompani et al., 2017) in mice indicated that a rather large number (≤ 91) of RGCs can converge onto single dLGN neurons. These numbers appear high compared to previous estimates but are not necessarily in conflict with functional findings, in particular if many of the structurally identified synapses had low weights (Chen et al., 2016), or were subject to later functional refinement, including synapse elimination and axon retraction, which occurs in mice between the ages of P9 and P60 (Chen and Regehr, 2000; Hong et al., 2014; Litvina and Chen, 2017). Convergence and divergence in dLGN has been hypothesized to underlie several functions, such as promotion of synchrony (Alonso et al., 1996; Usrey et al., 1998), enhancement of resolution (Alonso et al., 2006; Martinez et al., 2014), and robustness of feature detectors (Liang et al., 2018). These roles of convergence in retino-geniculate circuits seem different compared to those of convergence in thalamocortical circuits, where in mice, ~80 dLGN relay cells have been estimated to converge on V1 L4 simple cells with highly precise spatiotemporal offset to provide an origin for cortical direction selectivity (Lien and Scanziani, 2013, 2018).

Similarly, the estimated number of RGC types projecting to a single dLGN relay cell varies across studies and will likely differ depending on species (reviewed in Chen et al., 2016). In cats, where dLGN is organized into layers, paired recordings of RGCs (Usrey et al., 1999) or S potentials (Cleland et al., 1971) and dLGN relay cells have generally found an increase in connection probability with increasing similarity of RF properties. In particular, monosynaptic connections between RGCs and dLGN relay cells of opposite polarity seem rare (Cleland et al., 1971; Mastronarde, 1987, 1992; Usrey et al., 1999), indicating that, at least in cats, convergence of RGC types is low. The most direct estimates of convergence among RGC types to date has been obtained in mice, where single dLGN relay cell-

initiated mono-transsynaptic rabies virus tracing combined with morphological analysis of labeled RGCs revealed that some dLGN relay cells received input from RGCs of mostly one anatomical RGC type (relay mode), and others received input from up to nine different anatomical types (combination mode; Rompani et al., 2017). The presence of both relay- and combination-type retinogeniculate convergence in mouse thalamus seems consistent with recent ultrastructural (Morgan et al., 2016) and functional analyses (Liang et al., 2018) of thalamic RGC boutons, demonstrating both specificity and mixing: EM reconstructions of several thalamocortical dLGN cells and their retinal afferents revealed that two morphologically distinct RGC axon types made connections to different sets of target geniculate relay cells, as expected if these represented separate information channels. At the same time, these RGC types also converged onto an additional set of mixed-input thalamocortical neurons, consistent with convergence of types (Morgan et al., 2016). Similarly, a recent two-photon Ca^{2+} imaging study of RGC boutons in the shell region of dLGN has found that, at the scale of 2–6 μm , boutons from different RGCs often exhibited similar response preferences for one or several stimulus features, such as orientation, spatial frequency, and ON-OFF polarity (Liang et al., 2018). Critically, on this scale, RGC boutons have up to 50% probability of converging onto the same dLGN neuron (Liang et al., 2018). Hence, functional similarity of bouton clusters for multiple stimulus features might indicate convergence from single RGC types, and similarity in only one feature in the presence of mismatches in other feature preferences might indicate convergence from multiple RGC types (Liang et al., 2018). In line with these previous studies, our model of functional retinogeniculate convergence provides evidence for both specificity and mixing of RGC input types. At least for those dLGN neurons responding well to our stimulus set, it predicts that synaptic strength would be high for 1 or 2 types of functional RGC inputs and moderate for the remaining ones.

Our modeling results thus agree with the presence of both relay- and combination-mode functional convergence but critically extend the recent results to spatial scales larger than dendritic domains and to dLGN core. In addition, our results highlight the importance of considering connection weight in determining how many and which inputs are functionally relevant: with a liberal weight threshold, response prototypes from 3–8 RGC types are required to account for single-cell dLGN responses, compatible with the recent anatomical estimates from EM reconstructions and mono-transsynaptic rabies virus tracing (Morgan et al., 2016; Rompani et al., 2017). Raising the threshold to weights that likely have functionally relevant impact reduces the estimated number of RGC types providing inputs to mouse dLGN neurons to two, consistent with the large body of literature on functional convergence in the retino-geniculate pathway. It would be interesting to test these predictions on the identified connections in the published EM dataset from the Lichtman lab (Morgan et al., 2016), where ultrastructural features of the active zones, such as the volume of the postsynaptic densities and size of vesicle pools, may be assessed as indicators of synaptic strength.

Geniculate shell and core regions (Grubb and Thompson, 2004) differ in type and diversity of retinal projections (Huberman et al.,

2009; Kay et al., 2011; Kim et al., 2010; Rivlin-Etzion et al., 2011), as well as functional responses (Cruz-Martin et al., 2014; Piscopo et al., 2013). Although our dorsal approach for virus injection and electrophysiology traverses both shell and core, the volume difference between the regions suggests that our results are dominated by the dLGN core. Among our recorded population of dLGN neurons, we see a trend for higher DS in neurons recorded more dorsally (putative shell) versus ventrally (putative core region) within dLGN (Figure S6). We also observe a relative absence of DS for dLGN neurons with RFs covering the frontal space, further pointing to regional functional differences in dLGN representations. Interestingly, clusters of RGC terminals seem larger and more widespread in dLGN shell than core (Hammer et al., 2015), consistent with the possibility of more functional convergence in the shell. At the same time, retrograde transsynaptic tracing suggests a labeled-line communication for retinal direction selectivity running via the dLGN shell region to the upper layers of V1 (Cruz-Martin et al., 2014). Future studies will be required to explicitly compare and contrast the rules of retino-geniculate convergence between dLGN shell and core regions.

Considering its simplicity, our feedforward model of dLGN responses performed well. This is surprising, because the model does not consider known components of geniculate circuits, such as local inhibitory interneurons (Martinez et al., 2014), cortico-geniculate feedback (Sillito et al., 2006), and neuromodulatory influences (Antal et al., 2010; Fitzpatrick et al., 1989), and ignores transformations arising at the retino-geniculate synapse, such as postsynaptic summation (Carandini et al., 2007; Casti et al., 2008; Rathbun et al., 2010) and increases in contrast gain control (Alitto et al., 2018; Kaplan et al., 1987; Rathbun et al., 2016; Scholl et al., 2013). Potential explanations for the model's success—despite disregarding these essential circuits—include the spatial simplicity of our stimuli and the low-pass nature of our retinal signals. Indeed, local inhibition is known to increase push-pull mechanisms in dLGN, thereby enhancing temporal and spatial contrast (Hirsch et al., 2015; Martinez et al., 2014). These mechanisms will likely play a larger role in the processing of local or more complex stimuli, such as natural scenes (Lesica et al., 2006), than in the full-field stimulus used here. Similarly, performing the model predictions at the temporal resolution of Ca^{2+} signals smoothens over fine-grained differences in geniculate firing patterns, such as precise responses to high-frequency stimulation or bursting (Hirsch et al., 2015). In the future, it will be essential to extend our model by incorporating more of the geniculate circuit elements and to challenge the model predictions with more complex visual stimulation and signals of higher temporal resolution.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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● DATA AND SOFTWARE AVAILABILITY

SUPPLEMENTAL INFORMATION

Supplemental Information includes ten figures and one table and can be found with this article online at <https://doi.org/10.1016/j.neuron.2019.01.040>.

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and Virus Strains		
Cre-encoding Herpes-Simplex-Virus 1	MIT Vector Core, Cambridge, USA	hEFl α -cre - RN425
Experimental Models: Organisms/Strains		
Ai95(RCL-GCaMP6f)-D or Ai95D mice, B6;129S-Gt(ROSA) ^{tm95.1(CAG-GCaMP6f)Hze} /J	The Jackson Laboratory	IMSR Cat#JAX:024105; RRID: IMSR_JAX:024105
C57BL/6J mice	The Jackson Laboratory	RRID: IMSR_JAX:000664
Software and Algorithms		
QDSpy 0.77	developed and maintained by Thomas Euler	https://github.com/eulerlab/QDSpy
Python 3.4	https://www.python.org/	https://www.python.org/ ; RRID: SCR_008394
MATLAB R2017a	MathWorks	https://www.mathworks.com/products/matlab/ ; RRID: SCR_001622
IgorPro 6.3.7.2	Wavemetrics	https://www.wavemetrics.com/products/igorpro/igorpro.htm ; RRID: SCR_000325
SARFIA toolbox for IGOR Pro	Dorostkar et al., 2010	https://www.wavemetrics.com/project/SARFIA
Fiji/ImageJ	NIH	https://imagej.nih.gov/ij/ ; RRID: SCR_003070
Datajoint	Yatsenko et al., 2015	https://datajoint.github.io/ ; RRID: SCR_014543
EXPO visual display software	developed by Dr. Peter Lennie and maintained by Rob Dotson for the Center for Neural Science at New York University	https://sites.google.com/a/nyu.edu/expo/home
NManager software suite	Hazan et al., 2006	http://neurosuite.sourceforge.net/
KlustaKwik	Kenneth D. Harris	http://klustakwik.sourceforge.net/
Klusters	Hazan et al., 2006	http://neurosuite.sourceforge.net/
NNMF cross-validation	Alex H. Williams; Williams et al., 2018	https://gist.github.com/ahwillia/65d8f87fcd4bde3676d67b55c1a3954
Other		
Aladdin syringe pump	WPI Germany, Berlin, DE	AL-1000
Hamilton syringe	Hamilton Robotics, Reno, USA	Gastight (10 μ l); Model#1701;
Stereotactic frame	Neurostar, Tübingen, DE	Robot Stereotaxic
Closed loop temperature control system for small rodents	HD, Hugo Sachs Elektronik-Harvard Apparatus GmbH, March-Hugstetten, DE	Cat#50-7221-F
Dental drill motor	NSK	Volvere Vmax NE120 Model#VMAX35RV
Sharp micropipette	Science Products, Hofheim, DE	GB150F-8P
Ceramic filters	GE Healthcare, Buckinghamshire, UK	Anodisc #13, 0.2 μ m pore size
Vibratome	Thermo Fisher Scientific, Waltham, Massachusetts, USA	Microm HM 650 V
DAPI-containing mounting medium	Vector Laboratories Ltd, Peterborough, UK	Vectashield DAPI
Epi-fluorescent microscope	Zeiss, Oberkochen, DE	Zeiss Imager.Z1m
MOM-type two-photon microscope	designed by W. Denk, MPI, Martinsried; purchased from Sutter Instruments / Science Products, Hofheim, Germany; Euler et al., 2009	N/A
Mode-locked Ti:Sapphire laser	Newport Spectra-Physics, Darmstadt, Germany	MaiTai-HP DeepSee

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Fluorescence detection channel for GCaMP6f	AHF/Chroma Tübingen, Germany	HQ 510/84
Fluorescence detection channel for SR101	AHF/Chroma Tübingen, Germany	HQ 630/60, AHF
Water immersion objective	Zeiss, Oberkochen, Germany	W Plan-Apochromat 20x/1.0 DIC M27
DLP projector	Acer	K11
Band-pass-filtered light-emitting diodes 'green'	AHF/Chroma Tübingen, Germany	578 BP 10
Band-pass-filtered light-emitting diodes 'blue'	AHF/Chroma Tübingen, Germany	HC 405 BP 10
OptiBond FL primer and adhesive	Kerr dental, Rastatt, DE	Cat#35369
Dental cement	Ivoclar Vivadent, Ellwangen, DE	Tetric EvoFlow
Ground and reference screws	Bilaney	00-96 X 1/16 stainless steel screws
Kwik-Cast	WPI Germany, Berlin, DE	https://www.wpi-europe.com/products/laboratory-supplies/adhesives/kwik-cast.aspx
Gamma-corrected LCD screen	Samsung	Samsung SyncMaster 2233
32-channel edge silicon probes	Neuronexus, Ann Arbor, USA	A1x32Edge-5mm-20-177-A32
Arduino-type microcontroller	Arduino	http://www.arduino.cc/
Infrared light illumination	Allied Vision, Exton, USA	Guppy AVT camera
Red-shifted fluorescent lipophilic tracer	Thermo Fisher Scientific, Waltham, Massachusetts, USA	DiD
Blackrock microsystems amplifier	Blackrock Microsystems Europe GmbH, Hannover DE	128-Channel Neural signal processor

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to, and will be fulfilled by, the Lead Contact, Laura Busse (busse@bio.lmu.de).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

All procedures complied with the European Communities Council Directive 2010/63/EC and the German Law for Protection of Animals, and were approved by local authorities, following appropriate ethics review.

Functional characterization of dLGN-p RGCs

We used five mice of either sex aged 8 to 12 weeks of the Ai95D reporter line (B6; 129S-Gt(ROSA)26Sor^{tm95.1(CAG-GCaMP6f)Hze}/J; JAX 024105; RRID:IMSR_JAX:024 105). Ai95D features a floxed-STOP cassette preventing transcription of the genetically-encoded Ca²⁺ indicator GCaMP6f (Chen et al., 2013). Stereotactic injection of a Cre-encoding Herpes-Simplex-Virus 1 (hEFl α -cre, MIT Vector Core, Cambridge, USA) into the dLGN resulted in retrograde Cre-recombinase expression in dLGN-projecting (dLGN-p) RGCs, where Cre-recombinase, in turn, removed the LoxP sites and activated GCaMP6f expression.

Functional characterization of dLGN responses

We used 8- to 12-week-old wild-type mice (C57BL/6J; RRID: IMSR_JAX:000 664) of either sex.

METHOD DETAILS**Functional characterization of dLGN-p RGCs****Virus injection**

After induction of anesthesia using isoflurane (4% in oxygen), mice received isoflurane maintenance anesthesia (typically 1.2% in oxygen, adjusted to maintain a surgical plane of anesthesia as assessed by the absence of the pedal reflex), and were fixed in a stereotactic frame (Neurostar, Tübingen, DE). At the beginning of the surgical procedure, atropine (Atropine sulfate, 0.3 mg/kg, sc, Braun, Melsungen, DE) and analgesics (Buprenorphine, 0.1 mg/kg, sc, Bayer, Leverkusen, DE) were administered, and the eyes were protected with an eye ointment (Bepanthen, Bayer, Leverkusen, DE). The animal's temperature was kept constant at 37°C via a closed loop temperature control system for small rodents (HD, Hugo Sachs Elektronik-Harvard Apparatus GmbH, March-Hugstetten, DE).

After a midline scalp incision, a small hole was made with a dental drill (Sinco, Jengen, DE) over the dLGN located in the left hemisphere, 2.5 mm posterior to the bregma and 2.3 mm lateral from the midline. The virus was loaded in a sharp micropipette

(GB150F-8P, Science Products, inner tip diameter 20–25 μm , Hofheim, DE) connected through a 10 μl Hamilton syringe (Hamilton Robotics, Reno, USA) to an Aladdin syringe pump (AL-1000, WPI Germany, Berlin, DE). A volume of 20–40 nL of virus was injected at a depth of 2.7 mm. The pipette was left in place for an additional 5 min to allow for viral diffusion. Antibiotics (Baytril, 5 mg/kg, sc, Bayer, Leverkusen, DE) and a longer lasting analgesic (Carprofen, 5 mg/kg, sc, Rimadyl, Zoetis, Berlin, DE) were administered continuously for 3 days post-surgery. Two-photon Ca^{2+} imaging was carried out 3 weeks after viral injection.

Perfusion and retinal tissue preparation

Animals were housed under a standard 12 h day/night rhythm. Before perfusion and two-photon imaging, animals were dark-adapted for ≥ 1 h, and then deeply anaesthetized with a lethal dose of sodium pentobarbital (Narcoren, 400 mg/kg, injected intraperitoneally, Böhringer Ingelheim, Ingelheim, DE). When the animal reached the asphyxia stage and complete paralysis, the eyes were enucleated, and the mouse was transcardially perfused with 0.2 M sodium phosphate buffered saline (PBS), followed by 4% paraformaldehyde (PFA) solution in PBS. The brains were postfixed in PFA for 24 h at 4° and then stored in PBS.

The eyes were dissected in carboxygenated (95% O_2 , 5% CO_2) extracellular solution containing (in mM): 125 NaCl, 2.5 KCl, 2 CaCl_2 , 1 MgCl_2 , 1.25 NaH_2PO_4 , 26 NaHCO_3 , 20 glucose, and 0.5 L-glutamine (pH 7.4). The retina was extracted from the eyecup and flat-mounted onto a ceramic filter (Anodisc #13, 0.2 μm pore size, GE Healthcare, Buckinghamshire, UK) with the ganglion cell layer (GCL) facing up and transferred to the recording chamber of the microscope, where it was continuously perfused with carboxygenated solution at $\sim 36^\circ\text{C}$. In all experiments, ~ 0.1 μM Sulforhodamine-101 (SR101, Sigma, Steinheim, DE) was added to the extracellular solution to reveal blood vessels and any damaged cells in the red fluorescence channel of the microscope (Euler et al., 2009). All procedures were carried out under dim red (> 650 nm) illumination.

Histological reconstruction of injection sites

To verify the injection site within the dLGN, we used histological reconstructions. Brains were sliced for coronal sections (50 μm) using a vibratome (Microm HM 650 V, Thermo Fisher Scientific, Waltham, Massachusetts, USA) and mounted on glass slides with DAPI-containing mounting medium (Vectashield DAPI, Vector Laboratories Ltd, Peterborough, UK), which labels the cell nuclei, and coverslipped. Brain slices were inspected using a Zeiss Imager.Z1m epi-fluorescent microscope (Zeiss, Oberkochen, DE) for the expression of the retrogradely transported HSV-1, visualized by the expression of the GCaMP6f protein containing the enhanced green fluorescent protein (eGFP). The stereotactic slice coordinates were estimated based on Paxinos and Franklin (2008). We excluded one mouse from analysis of functional Ca^{2+} data because viral expression was spread beyond dLGN boundaries.

Two-photon Ca^{2+} imaging and light stimulation

We used a MOM-type two-photon microscope (designed by W. Denk, MPI, Martinsried; purchased from Sutter Instruments/Science Products, Hofheim, Germany). Design and procedures were described previously (Baden et al., 2016; Euler et al., 2009). In brief, the system was equipped with a mode-locked Ti:Sapphire laser (MaiTai-HP DeepSee, Newport Spectra-Physics, Darmstadt, Germany) tuned to 927 nm, two fluorescence detection channels for GCaMP6f (HQ 510/84, AHF/Chroma Tübingen, Germany) and SR101 (HQ 630/60, AHF), and a water immersion objective (W Plan-Apochromat 20x/1.0 DIC M27, Zeiss, Oberkochen, Germany). For image acquisition, we used custom-made software (ScanM, by M. Müller, MPI, Martinsried, and T. Euler) running under IGOR Pro 6.37 for Windows (Wavemetrics, Lake Oswego, OR, USA; RRID: SCR_000325), taking 64 $\times$ 64 pixel image sequences (7.8 frames/s) for activity scans or 512 $\times$ 512 pixel images for high-resolution morphology scans.

For light stimulation, we focused a DLP projector (K11, Acer) through the objective, fitted with band-pass-filtered light-emitting diodes (LEDs) (“green”, 578 BP 10; and “UV”, HC 405 BP 10, AHF/Chroma) to match the spectral sensitivity of mouse M- and S-opsins. LEDs were synchronized with the microscope’s scan retrace. Stimulator intensity (as photoisomerization rate, 10^3 P*/s/cone) was calibrated as described previously (Euler et al., 2009) to range from 0.6 and 0.7 (black image) to 18.8 and 20.3 for M- and S-opsins, respectively. An additional, steady illumination component of $\sim 10^4$ P*/s/cone was present during the recordings because of two-photon excitation of photopigments (for detailed discussion, see Baden et al., 2013; Euler et al., 2009). For all experiments, the tissue was kept at a constant mean stimulator intensity level for at least 15 s after the laser scanning started and before light stimuli were presented.

Four types of light stimuli were used (Figure 1D, top; Baden et al., 2016): (i) Full-field (800 $\times$ 600 μm) “chirp” stimuli consisting of a bright step and two sinusoidal intensity modulations, one with increasing frequency (0.5–8 Hz) and one with increasing contrast; (ii) 0.3 $\times$ 1 mm bright bar moving at 1 mm s $^{-1}$ in eight directions; (iii) alternating UV and green 3 s flashes; and (iv) binary dense noise (20 $\times$ 15 matrix with 40 μm pixel-side length; each pixel displayed an independent, balanced random sequence at 5 Hz for 5 min) for space-time receptive field mapping. All stimuli, except (iii), were achromatic, with matched photo-isomerization rates for mouse M- and S-opsins. Visual stimuli were presented using custom software (QDSpy, <https://github.com/eulerlab/QDSpy>).

Functional characterization of dLGN responses

Surgical procedures

After induction of anesthesia using isoflurane (4% in oxygen), mice received isoflurane maintenance anesthesia (typically 1.2% in oxygen, adjusted to maintain a surgical plane of anesthesia as assessed by the absence of the pedal reflex). At the beginning of the surgical procedure, atropine (Atropine sulfate, 0.3 mg/kg, sc, Braun, Melsungen, DE) and analgesics (Buprenorphine, 0.1 mg/kg, sc, Bayer, Leverkusen, DE) were administered, and the eyes were protected with an eye ointment (Bepanthen, Bayer, Leverkusen, DE). The animal’s temperature was kept constant at 37°C via a closed loop temperature control system for small rodents (HD, Hugo Sachs Elektronik-Harvard Apparatus GmbH, March-Hugstetten, DE).

After a midline scalp incision and skin removal, a drop of H_2O_2 (3%) was applied on the surface of the skull for the removal of tissue residues. A custom lightweight aluminum head post was placed on the posterior skull using OptiBond FL primer and adhesive (Kerr dental, Rastatt, DE) and Tetric EvoFlow dental cement (Ivoclar Vivadent, Ellwangen, DE). Miniature ground and reference screws (00-96 X 1/16 stainless steel screws, Bilaney) soldered to custom-made connector pins were placed bilaterally over the cerebellum. A well of dental cement was formed to hold the silicone elastomer sealant Kwik-Cast (WPI Germany, Berlin, DE) covering the skull. Antibiotics (Baytril, 5 mg/kg, sc, Bayer, Leverkusen, DE) and a longer lasting analgesic (Carprofen, 5 mg/kg, sc, Rimadyl, Zoetis, Berlin, DE) continued to be administered for 3 days post-surgery.

After recovery, animals were familiarized with a simulation of the experimental procedures in multiple training sessions until they were deemed comfortable with the conditions. Before experiments, a craniotomy (ca. 1 mm^2) was performed over dLGN (2.3 mm lateral to the midline and 2.5 mm posterior to bregma), which was re-sealed with Kwik-Cast (WPI Germany, Berlin, DE). Experiments started one day after craniotomy and were continued on consecutive days as long as electrophysiological signals remained of high quality.

In-vivo multisite extracellular recordings

Our experimental configuration for in-vivo recordings is described in detail in [Eriskien et al. \(2014\)](#). The mouse was head-fixed and could run freely on an air-suspended styrofoam ball while stimuli were presented on a gamma-corrected LCD screen (Samsung SyncMaster 2233). Extracellular neural signals were recorded 2.5 mm posterior from bregma and 2.3 mm lateral from midline through a small craniotomy window over dLGN with 32-channel edge silicon probes (Neuronexus, A1x32Edge-5mm-20-177- A32, Ann Arbor, USA). Neurons were verified as belonging to the dLGN based on the characteristic RF progression from top to bottom along the electrode shank ([Figure 3C](#)), the preference for high temporal frequencies, and a high prevalence of F1 responses to drifting gratings ([Grubb and Thompson, 2003](#); [Piscopo et al., 2013](#)). Ball movements were registered at 90 Hz by two optical mice connected to an Arduino-type microcontroller (<http://www.arduino.cc/>). Eye movements were monitored under infrared light illumination (Guppy AVT camera, frame rate 50 Hz, Allied Vision, Exton, USA).

Visual stimulation

We used custom software (EXPO, <https://sites.google.com/a/nyu.edu/expo/home>) to present visual stimuli on a gamma-calibrated liquid crystal display (LCD) monitor (Samsung SyncMaster 2233RZ; mean luminance 50 cd/m^2 , 60 Hz) at 25 cm distance to the animal's right eye. Four types of light stimuli were presented: (i) a "contrast stimulus" to measure the contrast response function, consisting of drifting sinusoidal gratings at a single orientation and 12 different randomly interleaved contrasts for 2 s with 5 s pauses between trials. (ii) the full-field chirp stimulus (see RGC section). (iii) a spatial-temporal-frequency-orientation (STFO) stimulus to capture preferred tuning properties of a large number of neurons simultaneously; the stimulus consisted of drifting sinusoidal gratings with 8 orientations, 6 temporal (0.5, 1, 2, 4, 8, 16 cycles/sec) and 2 spatial frequencies (0.5, 0.15 cycles/°) (443 of 814 cells, [Figure S7](#)). Trials were randomly interleaved and presented for 1 s with 0.1 s pauses between trials. The stimulus was shown at 100% contrast with the background at mean luminance. (iv) a sparse noise stimulus for receptive field mapping, consisting of white or black squares (5° visual angle) presented on a background of mean luminance (50 cd/m^2). Squares were flashed for 200 ms each at every position on a 12×12 square grid of 60 degrees.

Except for the sparse noise stimulus, all stimuli were presented in full-field mode. In experiments measuring tuning curves, a blank screen condition (mean luminance) was included to estimate the spontaneous firing rate.

Histological reconstruction of recording sites

To verify recording sites from dLGN, we used histological reconstructions. Before recording from the dLGN, electrodes were coated with a red-shifted fluorescent lipophilic tracer (DiD; Thermo Fisher Scientific, Waltham, Massachusetts, USA). After the last recording session, mice were transcardially perfused and the brain fixed in a 4% paraformaldehyde phosphate buffered saline (PBS) solution for 24 h and then stored in PBS. Brains were sliced for coronal sections ($50 \mu\text{m}$) using a vibratome (Microm HM 650 V, Thermo Fisher Scientific, Waltham, Massachusetts, USA) and mounted on glass slides with Vectashield DAPI (Vectashield DAPI, Vector Laboratories Ltd, Peterborough, UK), and coverslipped. Slices were inspected for DAPI and DiD presence using a Zeiss Imager.Z1m fluorescent microscope (Zeiss, Oberkochen, DE), and post-processed using FIJI/ImageJ (NIH; <https://imagej.nih.gov/ij/>; RRID: SCR_003070).

QUANTIFICATION AND STATISTICAL ANALYSIS

Functional characterization of dLGN-p RGCs

The data analysis was performed using IGOR Pro (Wavemetrics, Lake Oswego, OR, USA), MATLAB (The Mathworks, Natick, Massachusetts, USA) and Python (distribution by Anaconda Inc., Austin, TX; RRID:SCR_008394) using methods described previously ([Baden et al., 2016](#)).

Pre-processing

Regions of interest (ROIs) were manually drawn around the GCaMP6f-expressing somata in the recording fields. The Ca^{2+} traces for each ROI were extracted (as $\Delta F/F$) using the image analysis toolbox SARFIA for IGOR Pro ([Dorostkar et al., 2010](#)). A stimulus time marker embedded in the recorded data served to align the Ca^{2+} traces relative to the visual stimulus with a temporal precision of 2 ms. For this, Ca^{2+} traces were up-sampled to 512 Hz, and the timing for each ROI was corrected for sub-frame time-offsets related to the

scanning. The Ca^{2+} traces were then de-trended using a high-pass filtering above ~ 0.1 Hz and resampled to 7.8 Hz. For all stimuli except the dense noise (for RF mapping), the baseline was subtracted (median of first eight samples), median activity $r(t)$ across stimulus repetitions was computed (typically three to five repetitions) and normalized such that $\max_t(|r(t)|) = 1$.

Direction and orientation selectivity

To extract time course and directional tuning of the Ca^{2+} response to the moving bar stimulus, we performed a singular value decomposition (SVD) on the T by D normalized mean response matrix M (times samples by number of directions; $T = 32$; $D = 8$):

$$[U, S, V] = \text{svd}(M)$$

This procedure decomposes the response into a temporal component in the first column of U and a direction dependent component or tuning curve in the first column of V , such that the response matrix can be approximated as an outer product of the two:

$$M \approx S_{11} U_{:1} V_{:1}^T$$

An advantage of this procedure is that it does not require manual selection of time bins for computing direction tuning but extracts the direction-tuning curve given the varying temporal dynamics of different neurons.

To measure direction selectivity (DS) and its significance, we projected the tuning curve $V_{:1}$ on a complex exponential $\phi_k = \exp(i\alpha_k)$, where α_k is the direction of the k_{th} condition:

$$K = \phi^T V_{:1}$$

This is mathematically equivalent to computing the vector sum in the 2D plane or computing the power in the first Fourier component. We computed a DS index as the resulting vector length:

$$DSi = |K|$$

correcting for the direction spacing. We additionally assessed the statistical significance of direction tuning using a permutation test (Ecker et al., 2014). To this end, we created surrogate trials (that is, stimulus repetitions) by shuffling the trial labels (that is, destroying any relationship between condition and response), computed the tuning curve for each surrogate trial and projected it on the complex exponential ϕ . Carrying out the procedure 1,000 times generated a null distribution for K , assuming no direction tuning. We used the percentile of the true K as the p value for direction tuning (Baden et al., 2016).

Orientation selectivity (OS) was assessed in an analogous way. However, we used the complex exponential $\phi_k = \exp(2i\alpha_k)$, corresponding to the second Fourier component.

Signal deconvolution

Comparing neural activity measured with different methods (i.e., spikes versus Ca^{2+} signals; different Ca^{2+} indicators) is a non-trivial task. To assign dLGN-p RGCs to previously characterized RGC types, we needed account for the fact that the different Ca^{2+} indicators used (OGB-1 versus GCaMP6f) have different kinetics (Chen et al., 2013). We decided to convert both signals to a “common currency,” by deconvolving both signal types using Ca^{2+} kernels calculated for each indicator separately, using Ca^{2+} recordings of multiple ROIs ($n_{\text{OGB-1}} = 327$; $n_{\text{GCaMP6f}} = 19$) to the white noise stimulus, and averaging thresholded Ca^{2+} peak events ($> 80\%$ of the maximum normalized activity). The kernel area under the curve was normalized to 1 (Figure S3). These kernels were then used to deconvolve the respective calcium signal types.

Response quality index (Qi)

To measure how well a cell responded to a stimulus (local and full-field chirp, flashes), we computed the Qi, a signal-to-noise ratio, as a selection criterion:

$$Qi = \frac{\text{Var}[\langle \mathbf{C} \rangle_r]_t}{\langle \text{Var}[\mathbf{C}]_t \rangle_r}$$

where $\mathbf{C}$ is the T by R response matrix (time samples by stimulus repetitions), while $\langle \rangle_x$ and $\text{Var}[\]_x$ denote the mean and variance across the indicated dimension, respectively (Baden et al., 2016). If all trials are identical, such that the mean response is a perfect representative of the response, Qi is equal to 1. If all trials are completely random with fixed variance (so that the mean response is not informative about the individual trial responses at all), Qi is proportional to $1/R$. For further analysis, we used only cells that responded well to the chirp and/or to the moving bar stimulus ($Qi_{\text{chirp}} > 0.45$ or $Qi_{\text{DS}} > 0.6$; Baden et al. 2016). Of the original $n = 581$ ROIs, $n = 251$ ROIs passed this criterion (Figure S2).

Match index (Mi) for dLGN-p RGC type assignment

The pre-processed ROI traces of dLGN-p RGCs ($n = 251$) were assigned to the functional RGC population clusters reported by Baden et al. (2016), by identifying for each dLGN-p cell the cluster with the best matching response properties. After deconvolution with the respective Ca^{2+} kernel (see above), we calculated the linear correlation coefficients r between a dLGN-p cell's mean trace (over trials)

and all cluster mean traces (over all cells in an RGC population cluster) for the chirp stimulus (r_{chirp}) and the moving bar stimulus (r_{bar}). To combine the information about stimulus-specific correlations (r_{chirp} and r_{bar}), weighted by stimulus-specific cell response quality ($Q_{i_{chirp}}$ and $Q_{i_{bar}}$), we generated an overall match index (M_i) of each dLGN-p cell to all RGC population clusters:

$$M_i = \frac{Q_{i_{chirp}}}{Q_{i_{chirp}} + Q_{i_{bar}}} * r_{chirp} + \frac{Q_{i_{bar}}}{Q_{i_{chirp}} + Q_{i_{bar}}} * r_{bar}$$

Finally, each dLGN-p cell was assigned to the cluster with the highest M_i .

Cluster proportions in dLGN-p RGCs and RGC-all

We used a binomial test to assess the statistical significance of deviations of cluster cell numbers in the dLGN-p RGCs subpopulation compared to the RGC-all population. Starting with a total dLGN-p RGC cell number $n_{dLGN-p\ RGC} = 251$, for each dLGN-p RGC cluster i , we computed the test parameters k_i = “number of cells cluster i in dLGN-p RGC population,” and p_i = “proportion of cells in cluster i in the RGC-all population” ($n_{RGC-all} = 5024$), where p_i would be the expected proportion, or probability, of cells in that cluster if it were the same as in the RGC-all cluster. We then performed a two-tailed binomial test of obtaining a cell number deviation that extreme given the null hypothesis of equal cluster cell numbers in the two populations. After p value correction for false discovery rate (FDR, Benjamini-Hochberg), p values were considered significant at the significance level $\alpha = 0.01$.

For the $\log_2$ -ratio (% dLGN-p RGCs / % RGC population) plot, we used additive smoothing of the histogram, i.e., by adding one to each group’s cell count n per default and normalizing the group cell counts by $n+k$, where k is the number of clusters.

Linear feedforward model

For modeling dLGN responses as a linear combination of weighted RGC types, we used RGC-all cluster means (Baden et al., 2016) only of those clusters that were assigned at least two cells from our dLGN-p RGC data (Figure S2). The spike dataset recorded in dLGN was smoothed, down-sampled and convolved with an artificial OGB-1 kernel to allow for a direct comparison of the dLGN traces with the dLGN-p RGC Ca^{2+} responses. We then modeled dLGN responses as a linear combination of weighted RGC inputs (Figure S2). The weights were computed, using a linear-regression algorithm (*lsqlin*, MATLAB) with a non-negativity constraint, and then applied to their respective RGC population cluster responses, yielding the optimal prediction of the dLGN cell response. The non-negativity constraint is motivated by the non-negativity of feedforward excitatory connections between RGCs and dLGN. The model performance was cross-validated using repeated random sub-sampling with 1,000 repetitions. The trials were randomly shuffled and divided into a training set (50% trials) and a validation set (50% trials). Both sets thus contained unique trials with no duplicates. The training set was used to compute the weights and the performance of the model was evaluated on the validation set. In the end, a particular dLGN cell was modeled with the mean weights across all repeats, and the performance of the model represents the mean value across the repeats.

Functional characterization of dLGN responses

Data analysis was performed using MATLAB (The Mathworks, Natick, Massachusetts, USA; RRID:SCR_001622) and Python. Data were organized in a custom written schema using the relational database framework “DataJoint” (Yatsenko et al., 2015) (<https://datajoint.io>; RRID: SCR_014543).

Unit extraction and spike sorting

Wideband extracellular signals were digitized at 30 kHz (Blackrock microsystems, Blackrock Microsystems Europe GmbH, Hannover DE) and analyzed using the NManager software suite (Hazan et al., 2006), as described in detail in Erisken et al. (2014). The LFP was computed after downsampling to 1,250 Hz. To isolate single neurons from the linear arrays, we grouped neighboring channels into 5 equally sized “virtual octrodes” (8 channels per group with 2 channel overlap for 32 channel probes). Using a median-based automatic spike detection threshold multiplied by a factor of 1.5, spikes were extracted from the high-pass filtered continuous signal for each group separately. The first three principal components of each channel were used for semi-automatic isolation of single neurons with KlustaKwik. Clusters were manually refined with Klusters (Hazan et al., 2006). We assigned each unit to the contact with the largest waveform. Units were given a subjective quality score by the manual sorter, the firing rate, the cleanness of the refractory period, and the stability over time. To avoid duplication of neurons extracted from linear probe recordings, we computed cross-correlation histograms (CCHs, 1 ms bins) between neuron pairs from neighboring groups. Pairs for which the CCH’s zero-bin was three times larger than the mean of non-zero-bins were considered to be in conflict. For each conflicting pair, the cell with the best score was kept. Conflicts across pairs were resolved by collecting all possible sets of cells and by keeping the set with the best total score.

Chirp quality index

We used the chirp quality index as a selection criterion that determines whether a dLGN neuron was visually driven by the full-field chirp. We separated the stimulus into two segments (e.g., separated a 32 s stimulus into two 16 s segments), and computed the

average between-trial correlations (CCs) (responses binned at the stimulus frame rate) within segment and between segments. Only those cells that had significantly higher within-segment CCs ($Q_{i_{chirp}}$, $p < 0.01$, Wilcoxon rank sum test) and firing rate > 1 spike/s were considered to be visually responsive.

STFO correlation value

The STFO correlation value was a further unit selection criterion to assure neuronal response stability and to determine whether there were neurons that did not respond to the full-field chirp stimulus. We played the STFO stimulus directly before and after the chirp stimulus and performed linear regression analysis (Figure S7). To determine how well the model predicts the data, we computed a correlation value R :

$$R = \sqrt{1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

where $\hat{y}$ represents the calculated values of y and $\bar{y}$ is the mean of y .

For further analysis, we used only cells that responded well to the chirp stimulus ($Q_{i_{RF}} \geq 0.05$ and $Q_{i_{chirp}} \leq 0.001$) or had a high STFO correlation value ($R > 0.65$, Figure S2).

Receptive field mapping

Receptive fields were mapped by reverse correlating unit activity to the sparse noise stimulus and fitting the center of a two-dimensional ellipse / 2D-Gaussian for both ON- and OFF-fields:

$$f(x, y) = \frac{A}{2\pi ab} \exp\left(-\frac{x'^2}{2a^2} - \frac{y'^2}{2b^2}\right)$$

where A is the maximum amplitude, a and b are half-axes of the ellipse, and x' and y' are transformations of the stimulus coordinates x and y , considering the angle θ and the coordinates of the center (x_c, y_c) of the ellipse. For each contact, we computed a single RF center by averaging coordinates of the best-fit ON and OFF subfield (explained variance $> 70\%$).

Contrast response function

Contrast responses were fitted with a hyperbolic ratio function:

$$r_c = r_0 \frac{r_{max} * c^n}{(c_{50}^n + c^n)}$$

with baseline response r_0 , responsiveness r_{max} , semi-saturation contrast c_{50} , and exponent n .

Tuning

Orientation tuning curves were fitted with a sum of two Gaussians with peaks θ_{pref} and $\theta_{pref} - \pi$ of different amplitudes A_1 and A_2 but equal width σ , with a constant baseline r_0 . For each neuron, spatial and temporal frequency tuning curves were taken at its preferred direction; orientation and direction-tuning curves were taken at its optimal spatial and temporal frequencies.

Direction selectivity

Direction selectivity index (DSI) was calculated as the ratio of

$$DSI = \frac{r_{pref} - r_{opp}}{r_{pref} + r_{opp}}$$

where r_{pref} was the response at the preferred direction and r_{opp} was the response at the opposite direction. We additionally assessed the statistical significance of direction tuning using a permutation test (Ecker et al., 2014) as described above.

Orientation selectivity

Orientation selectivity index (OSI) was computed as:

$$OSI = \frac{r_{pref} - r_{ortho}}{r_{pref} + r_{ortho}}$$

where r_{pref} is the response to the preferred orientation and r_{ortho} is the response to the orthogonal orientation.

Non-negative matrix factorization (NNMF)

NNMF decomposes a matrix A into two low-rank non-negative matrices U and V^T , representing elementary components and their weights (Lee and Seung, 1999). In the case of our dLGN cell responses, this translates into finding a set of time-varying visual response components from which cell responses can be reconstructed as a weighted combination of those components. Given a

positive matrix A of size $N \times M$ (neuron $\times$ time) and a desired number of features k , the NNMF algorithm iteratively computes an approximation $A \sim UV^T$, where U and V^T are non-negative matrices with respective sizes $N \times k$ (neuron $\times$ weight) and $k \times M$ (component $\times$ time).

To determine the optimal number of components k , we applied NNMF cross-validation (Williams et al., 2018), also described by Alex H. Williams on <http://alexhwilliams.info/itsneuronalblog/2018/02/26/crossval/> and implemented on <https://gist.github.com/ahwillia/65d8f87fcd4bded3676d67b55c1a3954>. Analogous to cross-validation in supervised learning settings, NNMF cross-validation splits the data into a training and a test set in order to compare model performance on both sets and uses decreasing test set performance as a function of k as an indicator of model overfitting. In contrast to supervised settings, data entries cannot simply be held out for entire rows or columns as this would not allow to fit all parameters. Here, this problem was overcome by randomly holding out individual data entries from the data matrix and using the remaining data entries as training set. Formally, we define a binary matrix M , which acts as a mask over our data. The results in the following optimization problem:

$$\underset{U, V}{\text{minimize}} \quad \|M \circ (UV^T - A)\|_F^2$$

The optimization problem amounts to the low-rank matrix completion problem which was solved by alternating minimization of the above equation for U and V^T . The model error can then be evaluated on the held-out data points as follows:

$$\|(1 - M) \circ (UV^T - A)\|_F^2$$

We fitted NNMF models on the training data for ranks (= number of components k) 1 to 70, and compared the mean squared error (*MSE*) for the training versus the validation set. Due to the random model initialization for each run, this was done with $n_{\text{repeats}} = 200$. The optimal number of components was determined as the rank with the minimal average *MSE* over all runs $+ 5\%$.

The resulting components were organized into a hierarchical cluster tree by a single linkage algorithm using Euclidean distances and the Ward's minimum variance method. The results were plotted using the dendrogram function and the leaf order was optimized using the MATLAB function *optimalleaforder*.

Analysis of visual field coverage

To compute visual field coverage for dLGN recordings, we used the envelope of multi-unit activity (MUAe) to the sparse noise stimulus. To do so, we full-wave rectified the median-subtracted, high pass filtered signals, before low-pass filtering (200 Hz) and down-sampling to 2000 Hz (Supér and Roelfsema, 2005). To compute RF maps, we used average MUAe between 0 to 150 ms after stimulus onset (Figure S6A). To identify channels within the dLGN, we collapsed the MUAe RF map amplitudes for each channel across the x-dimension and range-normalized collapsed amplitudes. Channels with normalized amplitudes higher than an empirically set threshold (0.39) were included for further analysis. Non-detected channels located between detected channels were added. This procedure was evaluated against manually labeled ground truth data (using the Python Sloth GUI, <https://sloth.readthedocs.io/en/latest/>). We calculated RF centers as the location of the peaks of normalized amplitudes in x and y (Figures S6B and S6C). Since the distribution of DS cells has been suggested to vary between dLGN shell and core, we examined DS in a subpopulation of our recorded dLGN neurons with localized RFs and stable responses to the drifting gratings flanking the chirp stimulus ($n = 243$ cells). To test for the distribution of DSI as a function of RF location (Figure S6D), we extracted recorded units whose peak spike amplitude occurred at the identified RF-containing channels. To test for the distribution of DSIs as a function of dLGN depth, we compared DSIs of neurons in the top dLGN channel against the rest of the population (Figure S6E).

DATA AND SOFTWARE AVAILABILITY

The essential code and data to reproduce the analyses presented is available on <http://retinal-functomics.net/>.